Executive summary

- 102 software tests passing proves software consistency, not biological truth.
- DUD-E decoys are not experimentally verified inactive compounds.
- ChEMBL/DUD-E source separation may inflate performance.
- SHAP explains model behaviour, not biological causality.
- High benchmark performance does not prove therapeutic activity.
- Independent external validation has not yet been completed.
- Experimental validation is outside current scope.

Current benchmark design

See checksummed validation artifacts and scorecard for exact results.

Qualification: benchmark-supported with cautions.

What software validation proves

See checksummed validation artifacts and scorecard for exact results.

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What benchmark validation proves

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Source-bias analysis

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Adversarial validation

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Label-permutation testing

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Dummy and nearest-neighbour baselines

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Similarity-stratified performance

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Scaffold-stratified performance

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Descriptor-matched evaluation

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Source-balanced reweighting

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Activity-threshold sensitivity

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Decoy-subset sensitivity

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Calibration robustness

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Applicability domain

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Seed stability

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Explanation sanity checks

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External-validation status

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Validation scorecard

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Scientific limitations

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Recommended next validation step

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Reproducibility

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